

Preliminary analysis of effective assistance timing for iterative visual search tasks using gaze-based visual cognition estimation

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ABSTRACT

In this paper, focusing on whether a person has visually recognized a target (visual cognition, VC) in iterative visual-search tasks, we propose an efficient assistance method based on the VC. In the proposed method, we first estimate the participant's VC of the target in the previous task. We then determine the target for the next task based on the VC and start to guide the participant's attention to the target for the next task at the VC timing. By initiating the guidance from the timing of the previous target's VC, we can guide attention at an earlier time and achieve efficient attention guidance. The preliminary experimental results showed that VC-based assistance improves task performance.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

People collect information through various senses, such as sight and hearing, grasp the surrounding situation, and decide on appropriate actions based on such input. In general, human actions consist of three steps: *cognition*, *decision*, and *action* [1]. Focusing on only visual information, a task consists of several visual-search tasks (subtasks), which are repeated until the task's completion, as shown in Figure 2. In such a case, the decisions made during the previous subtask often affect the current subtask. Since the target to which a person needs to direct their attention changes according to the previous subtask's results, the target that requires attention in the current subtask needs to be changed appropriately. In this study, we attempt to develop an efficient method of guiding attention in such situations.

Many research approaches to the guidance of attention and gaze have been proposed [2]. In addition, since the field of view is limited when one wears a head-mounted display (HMD), researchers have also conducted many studies on gaze guidance in VR environments [3]. Most researchers have employed the feature of a person's visual acuity in peripheral vision being low, and they achieved attention and gaze guidance without being noticed by terminating visual changes before the targets reached the participants' central vision. However, these researchers did not consider iterative visual-search tasks. To assist with such tasks effectively, the time at which we start providing guidance is an important factor.

Therefore, we focus on visual cognition, specifically whether a person has recognized a target visually, and attempt to achieve efficient attention guidance at an earlier time based on visual-cognition estimation. To implement a visual-cognition estimation method, we conducted a preliminary experiment to analyze gaze behaviors when participants visually recognized the target.

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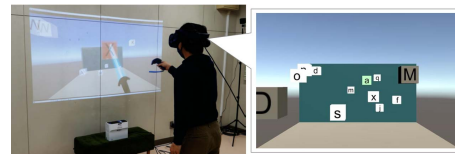


Figure 1: Experimental environment.

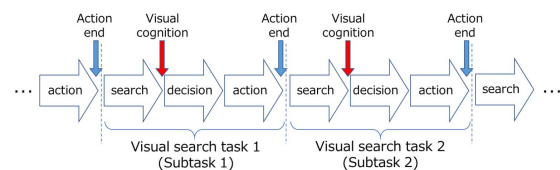


Figure 2: Flow of iterative visual search tasks

2 EXPERIMENTAL SETUP

2.1 Iterative visual-search tasks in experiment

We first describe the details of the experimental iterative visual-search tasks in a VR environment (Fig. 1). During the experiment, a large number of boxes with letters approached the participant. Here, the participant's tasks were to find the box with the designated character and to cut it using a VR controller.

As shown in Figure 1 right, most of the boxes display lowercase letters, but some of them display uppercase letters. These boxes are the possible 1st targets and are colored gray to make them easier to find. After a certain time (0.5 or 1.0 sec) from the 1st target's appearance, the boxes that display the lowercase letters appear. These boxes are possible 2nd targets. The first subtask is to find and select the 1st target among the candidates (the uppercase letters) and to cut the selected target. The second subtask is to find the 2nd target (the lowercase letters) that correspond to the 1st target's character and to cut it. We instructed the participants to complete these two subtasks iteratively.

2.2 Assistance types for iterative visual-search tasks

To evaluate the proposed method's effectiveness, we prepared the following four types of assistance. Here, "assistance" means coloring the 2nd target box green to make it easier to find.

- None: no assistance (Fig. 3 (a))
- All: assistance for all the possible 2nd targets as soon as they appear (Fig. 3 (b))
- AE (Action End): assistance for the 2nd target **after the end of the previous task** (cutting the 1st target) (Fig. 3 (c))
- VC (Visual Cognition): assistance for the 2nd target **from the visual-cognition timing** of the 1st target (Fig. 3 (d))

Note that Figs. 3 (c) and (d) appear the same, but the time at which we provide assistance differs between the AE and VC conditions.

2.3 Experimental environment

Figure 1 shows the experimental environment. The participant holds the VR controller. A light sword of $1005 \times 25 \times 25$ mm is attached to the controller in the VR environment. The participants cut the target boxes by touching the sword to the target. The experimental

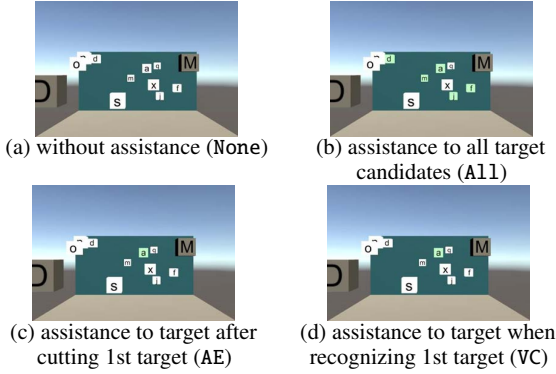


Figure 3: Assistance to 2nd target candidates

environment consisted of an HMD (HTC VIVE Pro Eye) and a VR controller (HTC VIVE controller 2018), which we created using Unity (2019.4.10f1) and Unity assets for VR (SteamVR, ViveSR). Before starting the experiment, we initialized the participant's position in the VR environment so that the center of the wall was in front of the participant.

The boxes with letters had sides of 25[cm]. These boxes emerged from the wall located 10[m] in front of the participant. They randomly appeared every 0.1 second from the area within a vertical range of ± 0.75 m and a horizontal range of ± 1.25 m from the center of the wall and approached the participant at 5 [m/s]. The boxes' horizontal and vertical positions did not change. We displayed nine letters (A, B, D, E, F, G, H, J, and M), whose uppercase and lowercase representations are easy to distinguish, on the target boxes.

First, the five possible 1st targets that display the uppercase letters appeared. After a certain time (0.5 or 1.0 seconds), the possible 2nd targets that display the lowercase letters appeared. The 1st candidates were colored gray because if these boxes were not colored (so they appeared the same as the other boxes), the possible 1st targets were too difficult to find.

2.4 Visual-cognition estimation

We estimated the visual cognition of the targets based on our previous research [4]. This study showed that the target's font size affected the minimum distance (angle) between the target and the gaze direction at the VC of the target in the VR environment. Based on the results, the visual cognition will be estimated when the angle between the participant's gaze and the target becomes smaller than the threshold. In the following experiment, we prepared three font sizes (s0: 16, s1: 32, s2: 48).

3 EXPERIMENT

Fifteen participants (21-23 years old, one female and 14 male) participated in the experiment. Before beginning the experiment, we explained it to the participants and obtained their oral informed consent for participation. Here, we only explained that we would assist with the tasks by coloring the boxes green. We did not explain the details of the assistance, such as the changing the assistance when the participants visually recognized the target in a certain condition. We also explained that the possible 1st targets were gray. After the explanation, we calibrated the gaze tracking of the HMD. Then, after the practice trials, the participant conducted the experiment.

The participant evaluated the four types of assistance shown in Fig. 3, three times each. The total number of experiments was 12. During the experiment, we recorded the gaze and head directions as well as the participant's cutting action. In the following analysis, we excluded one participant's results because he did not understand the meaning of "assistance" in this experiment.

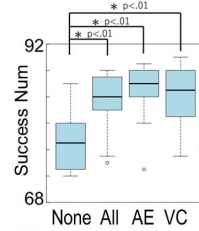


Figure 4: Number of task completions

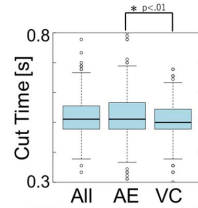


Figure 5: Duration between two-target cuttings

4 RESULTS

4.1 Visual-cognition estimation accuracy

First, we evaluated the accuracy of the visual-cognition estimation. Here, focusing on the five possible 1st targets, we considered that the 1st targets that were cut should be visually recognized (VC targets). Therefore, we determined whether the VC targets had actually been estimated as the VC. The estimation accuracy was 98.9% ($=1227/1241$). From these results, although the current method to estimate the visual cognition is very simple, we confirmed that the estimation accuracy was sufficient to evaluate the framework of the visual cognition-based assistance.

4.2 Effects on task completion

Next, we evaluated the effects of the assistance on the number of task completions. Figure 4 shows the comparison results. We conducted a one-factor repeated-measures analysis of variance (ANOVA) on the results. We confirmed that the assistance types can significantly affect the number of completions; therefore, we conducted multiple comparisons. These results are also shown in Fig. 4. Significant differences can be seen between cases without assistance (None) and with assistance (All, AE, VC); however, we could not confirm significant differences among the assistance conditions.

4.3 Effects on task completion time

We evaluated the effects of the assistance on the duration of the task completions. Here, we compared the elapsed time from the 1st target cutting to the 2nd target cutting in three conditions with assistance (All, AE, VC). Figure 5 shows the comparison results. We conducted a one-factor repeated-measures ANOVA on the results. These results are also shown in Fig. 5. The results indicate that although the difference is small, the elapsed time in the VC condition is significantly shorter than in the AE conditions.

5 CONCLUSION

In this paper, we proposed and evaluated the effectiveness of visual cognition-based assistance for the iterative visual-search tasks in a VR environment. The experimental results showed that the VC-based assistance improved task performance.

ACKNOWLEDGMENTS

This research was supported in part by JSPS KAKENHI Grant Numbers JP18K11383 and JP21K11968.

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